

Problem Set 1
Due: Mar. 1, 2021

1. Section 1.2 Problem 7.

A four-sided die is rolled repeatedly, until the first time (if ever) that an even number is obtained. What is the sample space for this experiment?

2. Section 1.2 Problem 8.

You enter a special kind of chess tournament, in which you play one game with each of three opponents, but you get to choose the order in which you play your opponents, knowing the probability of a win against each. You win the tournament if you win two games in a row, and you want to maximize the probability of winning. Show that it is optimal to play the weakest opponent second, and that the order of playing the other two opponents does not matter.

3. Let $\{A_i : i \in I\}$ be a collection of sets. Prove the "De Morgan's Law":

$$\left(\bigcup_i A_i\right)^c = \bigcap_i A_i^c, \quad \left(\bigcap_i A_i\right)^c = \bigcup_i A_i^c.$$

4. Six cups and saucers come in pairs: there are two cups and saucers which are red, two white, and two with stars on. If the cups are placed randomly onto the saucers (one each), find the probability that no cup is upon a saucer of the same pattern.